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COURSE 1.5 / COMPANION WORKBOOK

# Connect and segment the network

## Core routing, default-deny ACLs & internal time

Bring the eight VLANs online through a router the lab controls. Enforce default deny with a documented exception register, stand up the internal time service, and prove every permitted path and denial from the Windows side.

MATERIALS EDITION 1.0 / COURSE 1.5

Updated 7 October 2026 · sbd-connect-segment

NAME _____ / DATE _____

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Read each lesson alongside the interactive guide and the video. Use the record tables to write down what you actually observe — the workbook is your evidence, not the guide.

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Course review and sign-off

> **HOW TO USE IT** Each lesson page has the same shape: what you will learn, the build steps, the commands that verify them, and a record table. Fill the table from your own environment — never from the sample values.

# Before you start

This course connects the eight VLANs through a router the lab controls, enforces default-deny between them with a documented exception register, and stands up the internal time service. The router platform is your choice — the lessons describe what to configure and verify every result from the Windows side, so the checks work whatever appliance you pick. The network still has no path across the air gap.

## Prerequisites

- ☐ Complete courses 1.1–1.4: verified domain controllers, DNS, the hardened baseline with its scan cadence, and the two-tier PKI.
- ☐ Provision a new virtual machine for the layer-3 router (SBD-CORE01 in this sample) with one interface per VLAN, and a Windows Server 2025 VM for SBD-TIME01 at 10.0.20.40.
- ☐ Have the Inside the Network VLAN plan at hand: VLANs 10–80 on 10.0.X.0/24 subnets, with .1 reserved as each gateway in this sample.

## What this course builds

- Design and enforce default-deny segmentation with a justified exception register.
- Build the core router and prove permitted paths and logged denials per VLAN.
- Stand up SBD-TIME01 and verify the domain time hierarchy end to end.

## Systems in scope

| RECORD | YOUR OBSERVATION |
|---|---|
| SBD-CORE01 — name, address, OS build | |
| SBD-TIME01 — name, address, OS build | |

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LESSON 01 / ABOUT 25 MIN

# Design the segmentation and service paths

**WHY THIS MATTERS** A VLAN is a boundary only if something enforces it. Before touching the router, write the policy: default deny between VLANs, and an exception register where every permitted path names its source, destination, port, and reason.

## In this lesson you will

- Translate the Inside the Network VLAN plan into a default-deny segmentation policy.
- Draft the ACL exception register from the documented service paths.
- Record the gateway addressing and routing decisions before any device is built.

## Follow the build

1. List the eight VLANs with their subnets and purposes from the network plan. Reserve .1 in each subnet as the gateway for this sample, and record that no route crosses the air-gapped boundary.
2. Write the policy sentence first: traffic between VLANs is denied unless a register entry permits it. Segmentation alone does not establish enforcement — the access control lists do.
3. Draft the register's first entries from paths the built systems already need: every VLAN to the domain controllers for DNS, Kerberos, LDAP, SMB, and RPC; every system to SBD-TIME01 for time; the admin VLAN to the tiers it manages. One row per path: source, destination, service and port, justification, owner.
4. Mark the paths that do not exist yet (workstation pools, business services) as planned rows, and decide the change process: who may add a register entry, and what evidence each addition needs.

## Check your result

```
# Record the pre-routing state on each DC — today there is no gateway.
Get-NetIPConfiguration
Get-NetRoute -AddressFamily IPv4 | Select-Object DestinationPrefix, NextHop | Sort-Object
DestinationPrefix -Unique
```

**WHAT SUCCESS LOOKS LIKE** The register exists with every current path justified, the gateway plan is written, and the pre-routing state of both DCs is captured as the before picture.

## Your record

Evidence to keep: Keep the segmentation policy statement, the exception register with owners, and the captured pre-routing output.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| Evidence file names | |
| Open items or findings | |

## Think it through

What makes a VLAN an enforced boundary?

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> **REMEMBER** Default deny between VLANs; the register holds the exceptions. Every permitted path has a source, destination, port, reason, and owner. Gateways are planned (.1 per subnet) and the air gap stays unrouted.

Microsoft · Configure a firewall for Active Directory domains

Microsoft · Get-NetRoute

NIST · SP 800-53 Rev. 5 security controls

LESSON 02 / ABOUT 25 MIN

# Build the core router

> **WHY THIS MATTERS** The core is a virtual machine the lab controls: one interface per VLAN, the planned gateway address on each, and no default route anywhere — there is nowhere outside this network to go.

## In this lesson you will

- Deploy the layer-3 router VM with an interface in every VLAN.
- Assign the planned gateway addresses with routing enabled and no external route.
- Point the built systems at their gateways and prove basic reachability.

## Follow the build

1. Create the router VM (SBD-CORE01 in this sample) with one virtual interface attached to each of the eight VLAN networks. Record the platform and version you chose; the register, not the vendor syntax, is the course's deliverable.
2. Assign each interface its subnet's .1 address and enable routing between the interfaces. Do not configure a default route or any path toward the hypervisor's outside networks — the air gap is a routing decision too.
3. Set the gateway on the built systems: 10.0.10.1 on both domain controllers and 10.0.30.1 on the issuing CA. The offline root gets nothing; it stays off the network entirely.
4. From a host in another VLAN, prove the route: reach each gateway, then a DC address. Leave the ACLs permissive for this one lesson so routing and filtering failures cannot be confused — lesson 3 locks it down.

## Check your result

```
Get-NetIPConfiguration | Select-Object InterfaceAlias, IPv4Address, IPv4DefaultGateway
Test-NetConnection 10.0.10.1
Test-NetConnection 10.0.10.10 -TraceRoute
```

> **WHAT SUCCESS LOOKS LIKE** Each built system shows its planned gateway, the trace crosses the router in one hop, and the router holds no route toward anything outside the lab.

## Your record

Evidence to keep: Record the router platform and version, its interface and address table, the gateway changes on each system, and the trace output.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| Evidence file names | |
| Open items or findings | |

## Think it through

Why does the router get no default route?

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> **REMEMBER** One router interface per VLAN, .1 as each gateway, routing on. No default route — the air gap survives in the routing table. Reachability is proven before any filtering begins.

Microsoft · Test-NetConnection

Microsoft · Get-NetRoute

LESSON 03 / ABOUT 30 MIN

# Enforce default-deny with the register

**WHY THIS MATTERS** Now the policy becomes configuration: deny between VLANs by default, permit exactly the register's rows, and log what gets dropped so the denials are visible evidence, not silence.

## In this lesson you will

- Implement default-deny filtering between all VLANs.
- Permit the Active Directory, PKI, and time service paths from the register.
- Log denied traffic so enforcement produces evidence.

## Follow the build

1. Apply the inter-VLAN default-deny rule on the router, with logging for denied packets. The log of what was blocked is as much evidence as the list of what was allowed.
2. Permit the directory paths from every VLAN to VLAN 10 per Microsoft's Active Directory port documentation: DNS (53), Kerberos (88 and 464), LDAP (389/636), global catalog (3268/3269), SMB (445), and the RPC endpoint mapper with its dynamic range — the register row for RPC should name the range decision explicitly.
3. Permit NTP (123) from every VLAN to SBD-TIME01, and HTTP to the pki.sbd.test publication service in VLAN 30 so clients can fetch certificates and revocation lists.
4. Deny workstation-to-workstation traffic inside VLAN 80's future rows now, per the model's notes, and read back the full rule set against the register line by line — anything configured but not registered gets removed or registered, never ignored.

## Check your result

```
# From a test host outside VLAN 10: permitted paths answer, everything else fails.
Test-NetConnection SBD-DC01.sbd.test -Port 53
Test-NetConnection SBD-DC01.sbd.test -Port 88
Test-NetConnection SBD-DC01.sbd.test -Port 389
Test-NetConnection 10.0.10.10 -Port 3389   # expected: TcpTestSucceeded False
```

**WHAT SUCCESS LOOKS LIKE** Registered paths succeed, an unregistered port fails and appears in the router's deny log, and the rule set matches the register with no unexplained entries in either direction.

## Your record

Evidence to keep: Keep the exported rule set, the register cross-check, the permitted-path test output, and a captured deny-log entry for the blocked test.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| Differences from the plan | |
| Evidence file names | |
| Open items or findings | |

## Think it through

A rule exists on the router but not in the register. What happens?

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> **REMEMBER** Default deny is in force; the register's rows are the only openings. Denied traffic is logged — enforcement you can show, not assume. Rule set and register reconcile exactly, both directions.

Microsoft · Configure a firewall for Active Directory domains

Microsoft · Test-NetConnection

LESSON 04 / ABOUT 30 MIN

# Stand up SBD-TIME01 and the time hierarchy

> **WHY THIS MATTERS** Kerberos tolerates minutes of skew, not more, and an air-gapped network has no internet time to lean on. Build the internal reference, point the PDC emulator at it, and let the domain hierarchy carry time to everything else.

## In this lesson you will

- Configure SBD-TIME01 as the lab's internal reference clock.
- Point the PDC emulator at it and keep the rest on the domain hierarchy.
- Document the accuracy, fallback, and drift decisions the model calls for.

## Follow the build

1. Build SBD-TIME01 at 10.0.20.40 as a workgroup or domain-joined Windows Server 2025 host with the Windows Time service configured as an NTP server. In an air-gapped lab its clock is the reference by declaration — record the source decision (host clock, or an approved hardware source) with its expected accuracy and drift threshold.
2. On the domain controller holding the PDC emulator role, configure Windows Time to use SBD-TIME01 as its manual peer and mark it reliable. Only the PDC emulator points outward; every other domain member keeps the default domain-hierarchy sync.
3. Confirm the register's NTP row is active and the hardening GPOs from course 1.3 do not conflict with the time settings — if a STIG value and this design disagree, the deviation record from that course is where the decision lives.
4. Verify the chain end to end: TIME01 serving, the PDC emulator sourcing from it, the second DC sourcing from the domain hierarchy, and offsets within your recorded threshold.

## Check your result

```
# On the PDC emulator, then on the other DC.
w32tm /query /source
w32tm /query /status
w32tm /stripchart /computer:10.0.20.40 /samples:3 /dataonly
```

> **WHAT SUCCESS LOOKS LIKE** The PDC emulator reports SBD-TIME01 as its source, the other controller reports the domain hierarchy, and measured offsets sit inside the documented threshold.

## Your record

Evidence to keep: Record the time design decisions, both controllers' source and status output, the stripchart offsets, and the register row that permits NTP.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |

| RECORD | YOUR OBSERVATION |
| --- | --- |
| Differences from the plan | |
| Evidence file names | |
| Open items or findings | |

## Think it through

Which system syncs directly with SBD-TIME01?

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> **REMEMBER** TIME01 is the declared reference; its accuracy decisions are written down. Only the PDC emulator peers with it; the hierarchy carries the rest. Offsets are measured and inside the threshold, not assumed.

Microsoft · Windows Time service

Microsoft · Windows Time service tools and settings

LESSON 05 / ABOUT 25 MIN

# Prove the paths and the denials

**WHY THIS MATTERS** A segmented network is proven twice: every registered path works from where it is supposed to work, and the paths you never granted fail — visibly, in the log.

## In this lesson you will

- Test the permitted-path matrix from each VLAN against the register.
- Prove representative denials and capture their log entries.
- Confirm the directory still behaves correctly across the routed network.

## Follow the build

1. From a test host in each populated VLAN, run the register's permitted paths as a scripted matrix: directory ports to VLAN 10, NTP to TIME01, HTTP to pki.sbd.test. Record source VLAN, destination, port, and result for every row.
2. Run the deliberate failures: a port no register row permits, from a VLAN that should not reach it. Confirm the failure and find its entry in the router's deny log — the pair is the evidence.
3. Re-run the course 1.2 and 1.4 verification habits across the boundary: resolve the domain SRV records and fetch pki.sbd.test's certificate revocation list from outside VLANs 10 and 30.
4. File the matrix and the deny-log captures with the register. Any mismatch — a registered path failing or an unregistered path passing — is a finding to resolve before the course ends, not a note.

## Check your result

```
$targets = @(@('SBD-DC01.sbd.test',53),@('SBD-
DC01.sbd.test',88),@('10.0.20.40',123),@('pki.sbd.test',80))
foreach ($t in $targets) { Test-NetConnection $t[0] -Port $t[1] | Select-Object
ComputerName, RemotePort, TcpTestSucceeded }
Resolve-DnsName _ldap._tcp.dc._msdcs.sbd.test -Type SRV
```

**WHAT SUCCESS LOOKS LIKE** Every register row passes from its source VLAN, every probed unregistered path fails with a logged deny, and directory discovery works across the routed network.

## Your record

Evidence to keep: Keep the test matrix with timestamps, the deny-log captures, and the cross-VLAN SRV and CRL retrieval output.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| Evidence file names | |
| Open items or findings | |

## Think it through

What is the complete evidence for one denial?

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> **REMEMBER** Permits proven per VLAN; denials proven with their log entries. Directory discovery and CRL retrieval work across the boundary. Register and reality reconcile before the course moves on.

Microsoft · Configure a firewall for Active Directory domains

Microsoft · Test-NetConnection

LESSON 06 / ABOUT 15 MIN

# Delta scan the change and hand off

**WHY THIS MATTERS** Course 1.3's rule was simple: every role addition gets a scan. Two systems joined the network and every existing one gained a gateway — that is a change worth measuring.

## In this lesson you will

- Run the change-triggered scans on the touched Windows systems.
- Capture and hash the router configuration as its own baseline.
- Hand the register and evidence into the standing cadence.

## Follow the build

1. Rescan both domain controllers with the same benchmark and tool versions as course 1.3, and take SBD-TIME01's first baseline scan. Attribute any movement to this course's changes — the gateway and time settings are the expected deltas.
2. Export the router's configuration, hash the export, and store it with the evidence. The router has no SCAP benchmark in this lab, so its config export, register reconciliation, and deny log stand as its assessable record — say so explicitly in the evidence notes.
3. Add the new rows to the hardening course's exception and deviation record where the two overlap (time settings, any firewall interplay), and set the register's named owner and review date alongside the scan cadence.
4. Write the handoff line for the next course: the network now routes, filters, and keeps time — wave 2's PKI publication host and template administration build on exactly these paths.

## Check your result

```
Get-ChildItem C:\Evidence\connect-segment -Recurse | Select-Object Name, LastWriteTime
Get-FileHash -Algorithm SHA256 C:\Evidence\connect-segment\*
```

**WHAT SUCCESS LOOKS LIKE** The DC rescans show only the expected deltas, TIME01 has a dated baseline, the router export is hashed and reconciled with the register, and owners and review dates are set.

## Your record

Evidence to keep: Finish the course checklist: scan results, router export and hash, the reconciled register with owner and review date, and the handoff note.

| RECORD | YOUR OBSERVATION |
|---|---|
| Host(s) and time collected | |
| Observed result | |
| Differences from the plan | |

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| RECORD | YOUR OBSERVATION |
| --- | --- |
| Evidence file names | |
| Open items or findings | |

## Think it through

Why does this course end with scans at all?

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**REMEMBER** Change-triggered scans ran; deltas are attributed.  The router's record is its hashed export plus the register and deny log.  Owners, review dates, and the handoff to wave 2 are written.

NIST · SP 800-53 Rev. 5 security controls

# Course review and sign-off

Before moving on, confirm each lesson left evidence behind — and that what remains open is written down, not remembered.

## Lesson completion

| LESSON | EVIDENCE FILED | DATE |
|---|---|---|
| 01 Design the segmentation and service paths | ☐ yes | |
| 02 Build the core router | ☐ yes | |
| 03 Enforce default-deny with the register | ☐ yes | |
| 04 Stand up SBD-TIME01 and the time hierarchy | ☐ yes | |
| 05 Prove the paths and the denials | ☐ yes | |
| 06 Delta scan the change and hand off | ☐ yes | |

## Open items carried forward

## Next in the learning path

That completes course 1.5. Course 1.6 finishes the PKI services: the pki.sbd.test publication site on SBD-PKI-MGMT01 and certificate template administration.

> **KEEP IT HONEST** Lesson completion records learning activity; it is not a system authorization decision. Your observations, not the sample values, establish what happened in your environment.

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